MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR TRANSFORMER ROOM

DESIGN CALCULATION

AS BUILT

MD-216-49PR-EG-CV-CAL-2020

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	26-10-2023	AS BUILT	SHD	SGC	PRI	
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A01	15-10-2020	FIRST ISSUE	SHD	SGC	PRI	

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code

TABLE OF CONTENTS

		PAGE NO
DEFINITIONS and ABBREVIATIONS		3
1	INTENT	4
2	SCOPE	4
3	Design of Power Plant Transformer Room	5
	3.1 Load Calculation for Power Plant Transformer Room	5
	3.2 Design of Reinforced Concrete Members	7
	3.3 Design of Foundation mkd. F1	8
REFERENCES		11
ATTACHMENT 1	STAAD Input & Output File of Transformer Room	12
ATTACHMENT 2	Civil Foundation Input for Power Plant Station Transformer	59

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for Design Calculation of Power Plant Transformer Room

2. SCOPE

The document is prepared for Analysis & Structural Design Verification for Power Plant Transformer Room. Transformer Room is modelled in STAAD-Pro with relevant loads as per Design Input Drawing as attached.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
All the Reinforced Concrete members are designed in STAAD Pro using Factored Load Combinations from SBC 304:2018
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018
Net Safe Bearing Capacity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

3. Design of Power Plant Transformer Room

3.1 Load Calculation for Power Plant Transformer Room

All the Reinforced Concrete members are designed in STAAD using ACI Code.

L/C 3. Dead Load (DL) :

a) Roof Dead Load

i) 150 thk Hollow core slab =	2.52	kN/m^2
ii) Built-in Water Proofing system		
100 mm thk Polysterene Foam		
0.01kN/m^2 per 25.4mm thickness (100/25.4) =	0.04	kN/m^2
Bituminous smooth surface water proofing membrane =	0.07	kN/m^2
90mm thick concrete screed @0.38 kN/m2 for		
25 thickness (= 0.38*90/25)	1.37	kN/m^2
75mm thick concrete screed above HCS = 21 x0.075 =	1.6	kN/m^2
Sub Total =	**5.6**	**kN/m^2**
UDL on Roof beams = 5.6 x 2.75 x 0.5 =	**7.7**	**kN/m**

b) 355thk ground floor wall load = 20x3.3x0.355 = 23 kN/m (Clear height of wall = 3.3m)

L/C 4. Live Load (LR) :

a) Roof Live Load (non-accessible) = 0.5 kN/m^2 (as per Civil Structural Design Criteria)

UDL on Roof beams = **0.5 x5.5 x0.5 = 1.38 kN/m**

Wind load Calculation:

Wind load is calculated using the following parameters:

Basic Wind Speed	=	177	Km/hr	(for Non-Process Plant area as per Civil, Structural Design Criteria)
Exposure Category	=	D		(as per Civil, Structural Design Criteria)
Topographic factor, K_{zt}	=	1		(Section 6.4.3, SBC 301)
Importance Factor, I	=	1		(Section 6.5, SBC 301)
Velocity Pressure Co-efficient, K_z =		1.04		(Table 7.2-2 , SBC 301)
Wind Directionality Factor, K_d =		0.85		(Section 6.4.1, SBC 301)
Velocity pressure, $q_z = 0.0473 \times 10^{-3} \times K_z \times K_{zt} \times K_d \times V^2 \times I$				
	=	1.31	kN/m^2	

Enclosure Classificaton: Partially Enclosed Building (Section 6.2, SBC 301)

Internal Pressure Coefficient (GC_{pi}) = 0.5/-0.5 (Fig 7.2-1 , SBC 301)

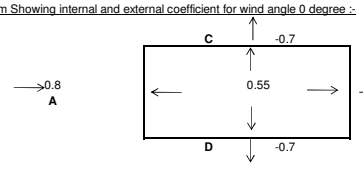
For External Pressure Coefficient (C_{pe}) Calculation :- (Fig 7.2-2 , SBC 301)

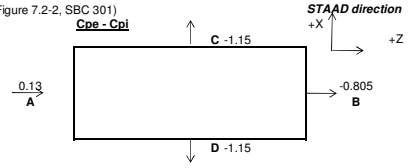
Height of Building, h =	5	m	h/L =	0.455
width of Building, B =	5.5	m		
Length of Building, L =	11	m	L/B =	2.000
Max. distance between columns =	5.5	m		

Internal pressure coefficient, GC_{pi} = +/- 0.55 (Fig 7.2-1 , SBC 301)

L/C 5. WIND IN +X DIRECTION (Cpi = +0.55) (Wind Angle = 0) (WX1)

Diagram Showing internal and external coefficient for wind angle 0 degree :- (Figure 7.2-2, SBC 301)

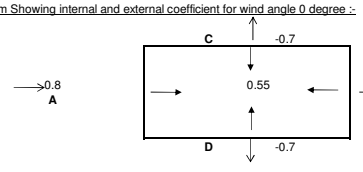


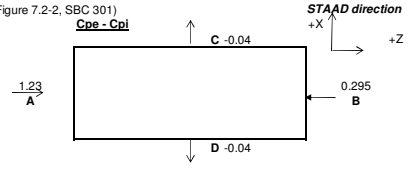


Area of Windward & Leeward faces each =	5 x 5.5 =	27.5	m2
Loads on Windward face A = qz*(Gcpe-Gcpi)*A =		4.68	kN
Loads on Leeward face B =		-29.00	kN
Area of other faces =	11 x 5 =	55	m2
Load on other faces =		-82.50	kN

L/C 6. WIND IN +X DIRECTION (Cpi = -0.55) (Wind Angle = 0) (WX2)

Diagram Showing internal and external coefficient for wind angle 0 degree :- (Figure 7.2-2, SBC 301)





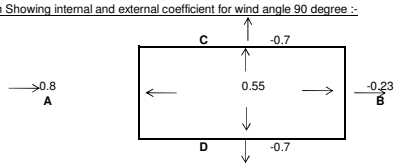
Area of Windward & Leeward faces each =	5 x 5.5 =	27.5	m2
Loads on Windward face A = qz*(Gcpe-Gcpi)*A =		44.31	kN
Loads on Leeward face B =		10.63	kN
Area of other faces =	11 x 5 =	55	m2
Load on other faces =		-3.24	kN

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Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

L/C 7. WIND IN +Z DIRECTION (Cpi = +0.55) (Wind Angle = 90) (WZ1)

Diagram Showing internal and external coefficient for wind angle 90 degree :-



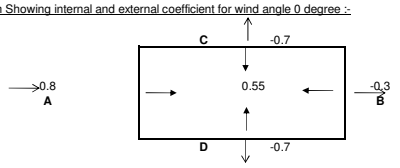
(Figure 7.2-2, SBC 301)



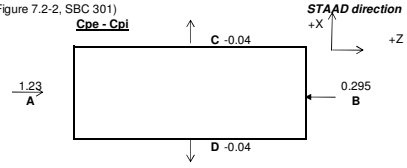
Area of Windward & Leeward faces each =	5 x 11 =	55	m2
Loads on Windward face A = qz*(Gcpe-Gcpi)*A =		9.37	kN
Loads on Leeward face B =		-53.71	kN
Area of other faces =	5 x 5.5 =	27.5	m2
Load on other faces =		-41.25	kN

L/C 8. WIND IN +Z DIRECTION (Cpi = -0.55) (Wind Angle = 90) (WZ2)

Diagram Showing internal and external coefficient for wind angle 0 degree :-



(Figure 7.2-2, SBC 301)



Area of Windward & Leeward faces each =	5 x 11 =	55	m2
Loads on Windward face A = qz*(Gcpe-Gcpi)*A =		88.62	kN
Loads on Leeward face B =		21.25	kN
Area of other faces =	5 x 5.5 =	27.5	m2
Load on other faces =		-1.62	kN

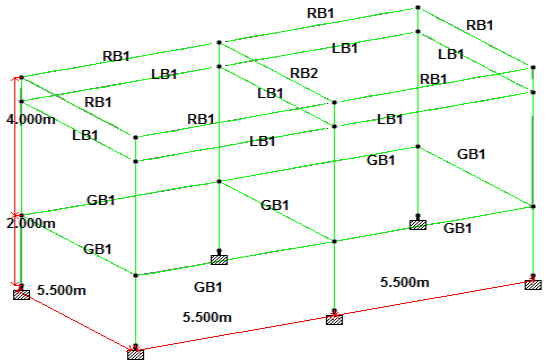
L/C 9. Temperature Load (TL) - Temperature Load applied for a temperature difference of 25^0.

L/C 1 & 2. Seismic Load (SL) - Seismic Load calculated & applied automatically in Staad as per IBC

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3.2 Design of Reinforced Concrete Members



3D MODEL OF TRANSFORMER ROOM REINFORCED CONCRETE FRAME IN STAAD-PRO

GRADE BEAM (GB1) - 375 WIDE X 500 DEEP
LINTEL BEAM (LB1) - 300 WIDE X 300 DEEP
ROOF BEAM (RB1 & RB2) - 300 WIDE X 450 DEEP
ALL COLUMNS ARE 500 X 500 MM^2 IN SIZE

All the Reinforced Concrete members except Foundation are designed in STAAD with the following factored load combinations as per SBC 304:2018

LOAD COMB 200 : (1.4 DL)

LOAD COMB 201: (1.4D+1.4T(+VE)+0.5LR)

LOAD COMB 202 : (1.4D+1.4T(-VE)+0.5LR)

LOAD COMB 203 : (1.2D+1.6WX1+0.5LR)

LOAD COMB 204 : (1.2D+1.6WX2+0.5LR)

LOAD COMB 205 : (1.2D+1.6WZ1+0.5LR)

LOAD COMB 206 : (1.2D+1.6WZ2+0.5LR)

LOAD COMB 207 : (1.2D+EZ)

LOAD COMB 208 : (1.2D+EX)

LOAD COMB 209 : (0.90D+1.6WX1)

LOAD COMB 210 : (0.90D+1.6WX2)

LOAD COMB 211: (0.90D+1.6WZ1)

LOAD COMB 212 : (0.90D+1.6WZ2)

LOAD COMB 213 : (0.90D+EZ)

LOAD COMB 214 : (0.90D+EX)

Please refer attached STAAD Output File for the following Beam & Column numbers for design

MEMBER SCHEDULE :

Member mkd.	Member No.	Size	Reinforcement Details
GB1	25, 26, 27, 28, 29, 30, 37	375X500	4 No. - D16 both at Top & Bottom
LB1	41,43,46,48,49,50,51	300X300	3 No. - D12 both at Top & Bottom
RB1 & RB2	1,2,6,7,11,13,38	300X450	3 No. - D16 both at Top & Bottom
COLUMNS	3,4,5,8,9, 10, 31, 32,33,34,35,36	500X500	Longitudinal Rebar 8 no. D20

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3.3 DESIGN OF FOUNDATION MKD. F1

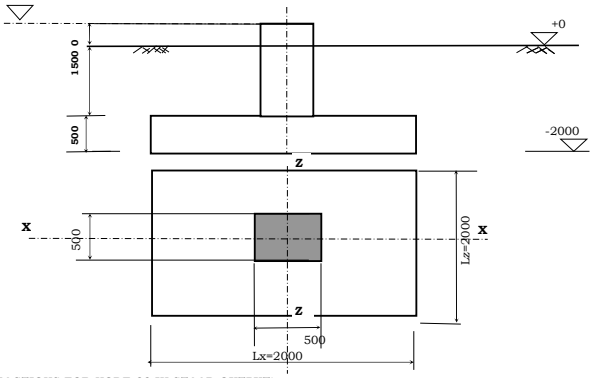
	About-X	About-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.50 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-2.00 m	
U/S level Base Plate :	0.00 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

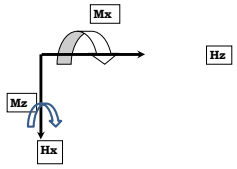
SYMBOLS AND NOTATIONS

Dead Load	D
live load	LL
Roof live load	LR
Wind Load	W
Seismic Load	E

Calculation of Weight of Foundation :

Weight of raft =	(2*2*0.5*2.5)	5.00T
Weight of soil =	((2*2)-(0.5*0.5))*1.5*1.8 =	10.13T
U/S Base Pl	Total =	15.13T





Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 415mm = 0.415m
Effective depth of slab along Z direction = 395mm = 0.395m
Clear projection along Z direction = 0.75m
Clear projection along X direction = 0.75m

(REFER SUPPORT REACTIONS FOR NODE 23 IN STAAD OUTPUT)

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
100 SLS: (DL)	2.37	25.44	2.63	1.55	-1.4	40.57	2.37	1.55	2.63	1.36
101 SLS: (DL+T)	8.64	26.04	6.21	5.50	-8.5	41.17	8.64	5.50	6.21	8.45
102 SLS: (D+LR)	2.35	26.15	2.61	1.54	-1.4	41.28	2.35	1.54	2.61	1.36
103 SLS: (D+0.75LR)	2.36	25.97	2.61	1.54	-1.4	41.10	2.36	1.54	2.61	1.36
104 SLS: (D+0.75*0.7EZ)	2.37	25.02	2.25	0.81	-1.4	40.15	2.37	0.81	2.25	1.36
105 SLS: (D+0.75*0.7EX)	2.03	25.09	2.63	1.55	-0.7	40.22	2.03	1.55	2.63	0.67
106 SLS: (D+0.75WX1+0.75LR)	2.07	25.75	2.67	1.61	-0.8	40.88	2.07	1.61	2.67	0.84
107 SLS: (D+0.75WX2+0.75LR)	1.99	25.74	2.62	1.55	-0.8	40.87	1.99	1.55	2.62	0.75
108 SLS: (0.6D+WX1)	1.04	14.97	1.66	1.02	-0.1	30.10	1.04	1.02	1.66	0.13
109 SLS: (0.6D+WX2)	0.93	14.95	1.58	0.93	0.0	30.08	0.93	0.93	1.58	0.01
110 SLS: (D+0.75WZ1+0.75LR)	2.44	25.45	1.98	0.43	-1.4	40.58	2.44	0.43	1.98	1.45
111 SLS: (D+0.75WZ2+0.75LR)	2.36	25.40	1.88	0.29	-1.4	40.53	2.36	0.29	1.88	1.36
112 SLS: (0.6D+WZ1)	1.53	14.56	0.73	-0.56	-0.9	29.69	1.53	0.56	0.73	0.93
113 SLS: (0.6D+WZ2)	1.43	14.50	0.60	-0.74	-0.8	29.63	1.43	0.74	0.60	0.82
114 SLS: (0.6D+0.7EX)	0.97	14.80	1.58	0.93	0.1	29.93	0.97	0.93	1.58	0.11
115 SLS: (0.6D+0.7EZ)	1.42	14.71	1.07	-0.06	-0.8	29.84	1.42	0.06	1.07	0.82

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Overturning & Sliding Check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
100 SLS: (DL)	40.57	2.37	1.36	2.63	1.55	29.74	26.16	6.85	6.17
101 SLS: (DL+T)	41.17	8.64	8.45	6.21	5.50	4.87	7.48	1.91	2.65
102 SLS: (D+LR)	41.28	2.35	1.36	2.61	1.54	30.38	26.79	7.03	6.33
103 SLS: (D+0.75LR)	41.10	2.36	1.36	2.61	1.54	30.22	26.64	6.97	6.30
104 SLS: (D+0.75*0.7EZ)	40.15	2.37	1.36	2.25	0.81	29.44	49.75	6.78	7.14
105 SLS: (D+0.75*0.7EX)	40.22	2.03	0.67	2.63	1.55	60.03	25.93	7.93	6.12
106 SLS: (D+0.75WX1+0.75LR)	40.88	2.07	0.84	2.67	1.61	48.67	25.42	7.90	6.12
107 SLS: (D+0.75WX2+0.75LR)	40.87	1.99	0.75	2.62	1.55	54.42	26.44	8.22	6.24
108 SLS: (0.6D+WX1)	30.10	1.04	0.13	1.66	1.02	240.80	29.60	11.58	7.25
109 SLS: (0.6D+WX2)	30.08	0.93	0.01	1.58	0.93	4297.14	32.21	12.94	7.62
110 SLS: (D+0.75WZ1+0.75LR)	40.58	2.44	1.45	1.98	0.43	28.04	94.81	6.65	8.20
111 SLS: (D+0.75WZ2+0.75LR)	40.53	2.36	1.36	1.88	0.29	29.74	138.33	6.87	8.62
112 SLS: (0.6D+WZ1)	29.69	1.53	0.93	0.73	0.56	31.79	53.40	7.76	16.27
113 SLS: (0.6D+WZ2)	29.63	1.43	0.82	0.60	0.74	36.00	40.20	8.29	19.75
114 SLS: (0.6D+0.7EX)	29.93	0.97	0.11	1.58	0.93	279.72	32.15	12.34	7.58
115 SLS: (0.6D+0.7EZ)	29.84	1.42	0.82	1.07	0.06	36.48	481.29	8.41	11.16

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	15-10-2020	31-10-2020	26-10-2023	Date	15-10-2020	31-10-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
200 ULS: (1.4 DL)	-3.32	35.61	-3.68	-2.17	1.91	50.740	-2.17	1.91
201 ULS: (1.4D+1.4T(+VE)+0.5LR)	-12.08	36.82	-8.68	-7.70	11.83	51.950	-7.70	11.83
202 ULS: (1.4D+1.4T(-VE)+0.5LR)	5.46	35.13	1.34	3.37	-8.02	50.260	3.37	-8.02
203 ULS: (1.2D+1.6WX1+0.5LR)	-2.22	30.41	-3.27	-2.00	0.53	45.540	-2.00	0.53
204 ULS: (1.2D+1.6WX2+0.5LR)	-2.05	30.39	-3.15	-1.86	0.34	45.520	-1.86	0.34
205 ULS: (1.2D+1.6WZ1+0.5LR)	-3.00	29.76	-1.79	0.52	1.82	44.890	0.52	1.82
206 ULS: (1.2D+1.6WZ2+0.5LR)	-2.84	29.66	-1.58	0.81	1.64	44.790	0.81	1.64
207 ULS: (1.2D+EZ)	-2.85	29.73	-2.43	-0.44	1.64	44.860	-0.44	1.64
208 ULS: (1.2D+EX)	-2.19	29.86	-3.15	-1.86	0.32	44.990	-1.86	0.32
209 ULS: (0.90D+1.6WX1)	-1.51	22.42	-2.49	-1.54	0.12	37.550	-1.54	0.12
210 ULS: (0.90D+1.6WX2)	-1.35	22.40	-2.37	-1.40	-0.07	37.530	-1.40	-0.07
211 ULS: (0.90D+1.6WZ1)	-2.30	21.77	-1.01	0.98	1.41	36.900	0.98	1.41
212 ULS: (0.90D+1.6WZ2)	-2.14	21.67	-0.80	1.27	1.24	36.800	1.27	1.24
213 ULS: (0.90D+EZ)	-2.13	22.10	-1.65	0.02	1.23	37.230	0.02	1.23
214 ULS: (0.90D+EX)	-1.48	22.23	-2.37	-1.40	-0.09	37.360	-1.40	-0.09

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

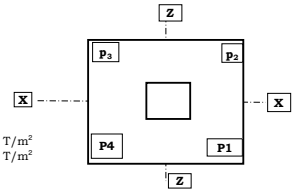
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28) For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m^2) V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-yh)	Remarks
100 SLS: (DL)	40.57	1.36	1.55	10.14	1.02	1.16	14.33	12.00	12.28	9.96	11.63	O.K
101 SLS: (DL+T)	41.17	8.45	5.50	10.29	6.34	4.13	22.76	14.51	10.08	1.83	20.06	O.K
102 SLS: (D+LR)	41.28	1.36	1.54	10.32	1.02	1.16	14.50	12.18	12.46	10.15	11.80	O.K
103 SLS: (D+0.75LR)	41.10	1.36	1.54	10.28	1.02	1.16	14.45	12.14	12.41	10.10	11.75	O.K
104 SLS: (D+0.75*0.7EZ)	40.15	1.36	0.81	10.04	1.02	0.61	13.67	12.46	11.62	10.41	10.97	O.K
105 SLS: (D+0.75*0.7EX)	40.22	0.67	1.55	10.06	0.50	1.16	13.72	11.39	12.72	10.39	11.02	O.K
106 SLS: (D+0.75WX1+0.75LR)	40.88	0.84	1.61	10.22	0.63	1.21	14.06	11.64	12.80	10.38	11.36	O.K
107 SLS: (D+0.75WX2+0.75LR)	40.87	0.75	1.55	10.22	0.56	1.16	13.94	11.62	12.81	10.49	11.24	O.K
108 SLS: (0.6D+WX1)	30.10	0.13	1.02	7.53	0.09	0.76	10.38	8.86	10.19	8.67	7.68	O.K
109 SLS: (0.6D+WX2)	30.08	0.01	0.93	7.52	0.01	0.70	10.23	8.82	10.22	8.81	7.53	O.K
110 SLS: (D+0.75WZ1+0.75LR)	40.58	1.45	0.43	10.15	1.09	0.32	13.55	12.91	11.38	10.74	10.85	O.K
111 SLS: (D+0.75WZ2+0.75LR)	40.53	1.36	0.29	10.13	1.02	0.22	13.37	12.94	11.33	10.89	10.67	O.K
112 SLS: (0.6D+WZ1)	29.69	0.93	0.56	7.42	0.70	0.42	10.54	9.71	9.14	8.31	7.84	O.K
113 SLS: (0.6D+WZ2)	29.63	0.82	0.74	7.41	0.62	0.55	10.58	9.47	9.34	8.24	7.88	O.K
114 SLS: (0.6D+0.7EX)	29.93	0.11	0.93	7.48	0.08	0.70	10.26	8.86	10.10	8.70	7.56	O.K
115 SLS: (0.6D+0.7EZ)	29.84	0.82	0.06	7.46	0.61	0.05	10.12	10.03	8.89	8.80	7.42	O.K

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

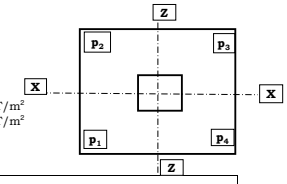
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 53.6 T/m^2 For WL/SL case = 66.1 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-yh)
200 ULS: (1.4 DL)	50.7	1.91	-2.17	12.69	1.43	1.63	17.75	14.49	14.88	11.62	14.15
201 ULS: (1.4D+1.4T(+VE)+0.5LR	52.0	11.83	-7.70	12.99	8.87	5.77	29.63	16.09	9.89	-1.66	26.03
202 ULS: (1.4D+1.4T(-VE)+0.5LR)	50.3	-8.02	3.37	12.57	6.01	2.52	23.10	16.05	9.08	4.03	19.50
203 ULS: (1.2D+1.6WX1+0.5LR)	45.5	0.53	-2.00	11.39	0.39	1.50	15.28	10.28	12.49	9.50	11.68
204 ULS: (1.2D+1.6WX2+0.5LR)	45.5	0.34	-1.86	11.38	0.25	1.40	15.03	10.24	12.52	9.73	11.43
205 ULS: (1.2D+1.6WZ1+0.5LR)	44.9	1.82	0.52	11.22	1.36	0.39	14.98	12.20	10.25	9.47	11.38
206 ULS: (1.2D+1.6WZ2+0.5LR)	44.8	1.64	0.81	11.20	1.23	0.61	15.04	11.82	10.58	9.36	11.44
207 ULS: (1.2D+EZ)	44.9	1.64	-0.44	11.22	1.23	0.33	14.78	12.11	10.32	9.66	11.18
208 ULS: (1.2D+EX)	45.0	0.32	-1.86	11.25	0.24	1.40	14.88	10.09	12.41	9.62	11.28
209 ULS: (0.90D+1.6WX1)	37.6	0.12	-1.54	9.39	0.09	1.15	12.63	8.33	10.45	8.15	9.03
210 ULS: (0.90D+1.6WX2)	37.5	-0.07	-1.40	9.38	0.05	1.05	12.49	8.38	10.38	8.28	8.89
211 ULS: (0.90D+1.6WZ1)	36.9	1.41	0.98	9.23	1.06	0.74	13.02	9.55	8.90	7.43	9.42
212 ULS: (0.90D+1.6WZ2)	36.8	1.24	1.27	9.20	0.93	0.95	13.08	9.17	9.23	7.32	9.48
213 ULS: (0.90D+EZ)	37.2	1.23	0.02	9.31	0.92	0.02	12.24	10.21	8.40	8.37	8.64
214 ULS: (0.90D+EX)	37.4	-0.09	-1.40	9.34	0.07	1.05	12.46	8.36	10.32	8.22	8.86

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	15-10-2020	31-10-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	15-10-2020	31-10-2020	26-10-2023
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Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.5+2*0.5*0.395)+(0.5+2*0.5*0.395))*2

= 3.58 m

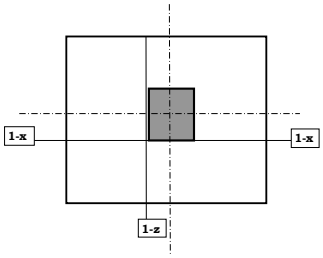
Punching Area= ((0.5+2*0.5*0.395)*(0.5+2*0.5*0.395))

= 0.8 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*2.5+1.5*1.8)

= 3.95 T/m^2

Area of footing, A_f= 4.00 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 83.30 T

Strength of Concrete in shear Vc1 = 302.43 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 447.12 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 278.86 T **Table 22.6.5.2_ (c)**

Design Vc= 278.86 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (83.3*10000)/(3580*395)

= 0.59 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
200 ULS: (1.4 DL)	14.15	3.98	4.74	3.98	5.02	45.28	0.32	OK	0.74	0.127	OK	0.26
201 ULS: (1.4D+1.4T(+VE)+0.5LR	26.03	7.32	8.72	7.32	9.24	83.30	0.59	OK	0.74	0.234	OK	0.47
202 ULS: (1.4D+1.4T(-VE)+0.5LR)	19.50	5.48	6.53	5.48	6.92	62.40	0.44	OK	0.74	0.175	OK	0.35
203 ULS: (1.2D+1.6WX1+0.5LR)	11.68	3.29	3.91	3.29	4.15	37.38	0.26	OK	0.74	0.105	OK	0.21
204 ULS: (1.2D+1.6WX2+0.5LR)	11.43	3.21	3.83	3.21	4.06	36.58	0.26	OK	0.74	0.103	OK	0.21
205 ULS: (1.2D+1.6WZ1+0.5LR)	11.38	3.20	3.81	3.20	4.04	36.42	0.26	OK	0.74	0.102	OK	0.21
206 ULS: (1.2D+1.6WZ2+0.5LR)	11.44	3.22	3.83	3.22	4.06	36.61	0.26	OK	0.74	0.103	OK	0.21
207 ULS: (1.2D+EZ)	11.18	3.14	3.75	3.14	3.97	35.78	0.25	OK	0.74	0.1	OK	0.20
208 ULS: (1.2D+EX)	11.28	3.17	3.78	3.17	4.00	36.10	0.26	OK	0.74	0.101	OK	0.20
209 ULS: (0.90D+1.6WX1)	9.03	2.54	3.03	2.54	3.21	28.90	0.2	OK	0.74	0.081	OK	0.16
210 ULS: (0.90D+1.6WX2)	8.89	2.50	2.98	2.50	3.16	28.45	0.2	OK	0.74	0.08	OK	0.16
211 ULS: (0.90D+1.6WZ1)	9.42	2.65	3.16	2.65	3.34	30.14	0.21	OK	0.74	0.085	OK	0.17
212 ULS: (0.90D+1.6WZ2)	9.48	2.67	3.18	2.67	3.37	30.34	0.21	OK	0.74	0.085	OK	0.17
213 ULS: (0.90D+EZ)	8.64	2.43	2.89	2.43	3.07	27.65	0.2	OK	0.74	0.078	OK	0.16
214 ULS: (0.90D+EX)	8.86	2.49	2.97	2.49	3.15	28.35	0.2	OK	0.74	0.08	OK	0.16

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
200 ULS: (1.4 DL)	0.34	0.001	415
201 ULS: (1.4D+1.4T(+VE)+0.5LR	0.63	0.004	1660
202 ULS: (1.4D+1.4T(-VE)+0.5LR)	0.47	0.001	415
203 ULS: (1.2D+1.6WX1+0.5LR)	0.28	0.001	415
204 ULS: (1.2D+1.6WX2+0.5LR)	0.27	0.001	415
205 ULS: (1.2D+1.6WZ1+0.5LR)	0.27	0.001	415
206 ULS: (1.2D+1.6WZ2+0.5LR)	0.27	0.001	415
207 ULS: (1.2D+EZ)	0.27	0.001	415
208 ULS: (1.2D+EX)	0.27	0.001	415
209 ULS: (0.90D+1.6WX1)	0.22	0.001	415
210 ULS: (0.90D+1.6WX2)	0.21	0.001	415
211 ULS: (0.90D+1.6WZ1)	0.23	0.001	415
212 ULS: (0.90D+1.6WZ2)	0.23	0.001	415
213 ULS: (0.90D+EZ)	0.21	0.001	415
214 ULS: (0.90D+EX)	0.21	0.001	415

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.
At bottom	7.32	0.47	0.63	0.004	1660.00	747.00	20ϕT	189.25	300	150
At top	0.00	0.00	0.00	0.0018	711.00	711.00	12ϕT	159.07	300	150

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**20ϕT**	**150**	c/c	At Top	**12ϕT**	**150**	c/c
Along Z Direction :	At Bottom	**20ϕT**	**150**	c/c	At Top	**12ϕT**	**150**	c/c

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Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

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Date	15-10-2020	31-10-2020	26-10-2023				Date	15-10-2020	31-10-2020	26-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 1 STAAD Input & Output File of Transformer Room

```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    OCT 15, 2020                   *
*           Time=    12:37:10                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: Transformer Room1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 18-OCT-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 -3 0 -1.3; 2 2.5 0 -1.3; 3 8 0 -1.3; 4 -3 4.99961 -1.3; 5 2.5 5 -1.3
     9. 6 8 4.99961 -1.3; 7 -3 0 4.2; 8 2.5 0 4.2; 9 8 0 4.2; 10 -3 4.99961 4.2
    10. 11 2.5 5 4.2; 12 8 5 4.2; 23 -3 -2 -1.3; 24 2.5 -2 -1.3; 25 8 -2 -1.3
    11. 26 -3 -2 4.2; 27 2.5 -2 4.2; 28 8 -2 4.2; 29 -3 4.29961 -1.3; 30 2.5 4.3 -1.3
    12. 31 8 4.29961 -1.3; 32 -3 4.29961 4.2; 33 2.5 4.3 4.2; 34 8 4.3 4.2
    13. MEMBER INCIDENCES
    14. 1 4 5; 2 5 6; 3 1 29; 4 2 30; 5 3 31; 6 10 11; 7 11 12; 8 7 32; 9 8 33
    15. 10 9 34; 11 4 10; 13 6 12; 25 1 2; 26 2 3; 27 3 9; 28 9 8; 29 8 7; 30 7 1
    16. 31 1 23; 32 2 24; 33 3 25; 34 7 26; 35 8 27; 36 9 28; 37 2 8; 38 5 11; 39 29 4
    17. 40 30 5; 41 29 30; 42 31 6; 43 30 31; 44 32 10; 45 33 11; 46 32 33; 47 34 12
    18. 48 33 34; 49 29 32; 50 31 34; 51 30 33
    19. DEFINE MATERIAL START
    20. ISOTROPIC CONCRETE
    21. E 2.17184E+007
    22. POISSON 0.17
    23. DENSITY 23.6158
    24. ALPHA 5E-006
    25. DAMP 0.05
    26. TYPE CONCRETE
    27. STRENGTH FCU 27578.9
    28. END DEFINE MATERIAL
    29. MEMBER PROPERTY AMERICAN
    30. 3 5 8 10 31 33 34 36 39 42 44 47 PRIS YD 0.5 ZD 0.5
    31. 1 2 6 7 PRIS YD 0.45 ZD 0.3
    32. MEMBER PROPERTY AMERICAN
    33. 27 30 37 PRIS YD 0.5 ZD 0.375
    34. MEMBER PROPERTY AMERICAN
    35. 11 13 38 PRIS YD 0.45 ZD 0.3
    36. MEMBER PROPERTY AMERICAN
    37. 4 9 32 35 40 45 PRIS YD 0.5 ZD 0.5
    38. MEMBER PROPERTY AMERICAN
```

```
39. 25 26 28 29 PRIS YD 0.5 ZD 0.375
40. MEMBER PROPERTY AMERICAN
41. 41 43 46 48 TO 51 PRIS YD 0.3 ZD 0.3
42. CONSTANTS
43. BETA 90 MEMB 4 9 32 35 40 45
44. MATERIAL CONCRETE ALL
45. SUPPORTS
46. 23 TO 28 FIXED
47. ******************************************************************************
48. MEMBER OFFSET
49. 2 7 43 48 START 0.2 0 0
50. 2 7 43 48 END -0.125 0 0
51. 1 6 41 46 START 0.125 0 0
52. 1 6 41 46 END -0.2 0 0
53. 11 13 49 50 START 0 0 0.2
54. 11 13 49 50 END 0 0 -0.2
55. DEFINE IBC 2003
56. SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
57. SELFWEIGHT 1
58. MEMBER WEIGHT
59. 27 30 UNI 10
60. MEMBER WEIGHT
61. 1 2 6 7 41 43 46 48 UNI 8.9
62. DEFINE WIND LOAD
```

***** NOTE: If any floor diaphragm is present in the model Wind Load definition should be defined after Floor Diaphragm definition. Otherwise wind load generation may be unsuccessful during analysis.**

```
63. TYPE 1 WIND X WINDWARD
64. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
65. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
66. 1 4.000 M 14.000 M 5.000 M 2.000 0.010 0 -
67. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
68. 0 0 0 0.900 0.800 -0.550
69. !> END GENERATED DATA BLOCK
70. INT 1.6486 1.6486 HEIG  0  4.572
71. TYPE 2 WIND X LEEWARD
72. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
73. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
74. 1 4.000 M 14.000 M 5.000 M 2.000 0.010 1 -
75. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
76. 0 0 0 0.900 -0.200 0.550
77. !> END GENERATED DATA BLOCK
78. INT -0.947728 -0.947728 HEIG  0  4
79. TYPE 3 SIDEWARD X
80. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
81. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
82. 1 4.000 M 15.000 M 5.000 M 2.000 0.010 2 -
83. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
84. 0 0 0 0.900 -0.700 0.550
85. !> END GENERATED DATA BLOCK
86. INT -1.53179 -1.53179 HEIG  0  4
87. TYPE 4 WIND Z WINDWARD
88. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
89. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
```

```
 90. 1 4.000 M 5.000 M 15.000 M 2.000 0.010 0 -
 91. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
 92. 0 0 0 0.886 0.800 -0.550
 93. !> END GENERATED DATA BLOCK
 94. INT 1.63434 1.63434 HEIG  0  4.572
 95. TYPE 5 WIND Z LEEWARD
 96. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
 97. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
 98. 1 4.000 M 5.000 M 15.000 M 2.000 0.010 1 -
 99. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
100. 0 0 0 0.886 -0.500 0.550
101. !> END GENERATED DATA BLOCK
102. INT -1.28925 -1.28925 HEIG  0  4
103. TYPE 6 SIDEWARD Z
104. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
105. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
106. 1 4.000 M 5.000 M 15.000 M 2.000 0.010 2 -
107. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
108. 0 0 0 0.886 -0.700 0.550
109. !> END GENERATED DATA BLOCK
110. INT -1.51931 -1.51931 HEIG  0  4
111. LOAD 1 LOADTYPE SEISMIC  TITLE EQK LOAD Z
112. IBC LOAD Z 1
113. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**24**	**NUMBER OF MEMBERS**	**39**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 18/ 6/ 42 DOF
TOTAL PRIMARY LOAD CASES = 1, TOTAL DEGREES OF FREEDOM = 108
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 5 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.1/ 160014.6 MB

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.016 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.269 T =  0.324 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.324                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.040 X      1090.31   *
*                       =        43.10  KN                 *
************************************************************
```

```
114. CHANGE
115. LOAD 2 LOADTYPE SEISMIC  TITLE EQK LOAD X
116. IBC LOAD X 1
117. PERFORM ANALYSIS
```

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.016 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.269 T =  0.297 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.297                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.042 X      1090.31   *
*                       =        46.23  KN                 *
************************************************************
```

```
118. CHANGE
119. LOAD 3 LOADTYPE DEAD  TITLE DL
120. SELFWEIGHT Y -1
121. MEMBER LOAD
122. 25 TO 30 37 UNI GY -23
123. MEMBER LOAD
124. 1 2 6 7 11 13 UNI GY -7.7
125. 38 UNI GY -15.5
126. *11 UMOM GZ 1
127. *13 UMOM GZ -1
128. *1 2 UMOM GX -1
129. *6 7 UMOM GX 1
130. LOAD 4 LOADTYPE LIVE  TITLE LL
131. MEMBER LOAD
```

```
132. 1 2 6 7 11 13 38 UNI GY -1.4
133. *11 UMOM GZ 0.1
134. *13 UMOM GZ -0.1
135. *1 2 UMOM GX -0.1
136. *6 7 UMOM GX 0.1
137. LOAD 5 LOADTYPE WIND  TITLE WL+X
138. JOINT LOAD
139. 1 4 7 10 FX 1
140. 3 6 9 12 FX 7.3
141. 1 TO 6 FZ -14
142. 7 TO 12 FZ 14
143. LOAD 6 LOADTYPE WIND  TITLE WL-X
144. JOINT LOAD
145. 1 4 7 10 FX 11
146. 3 6 9 12 FX -3
147. 1 TO 6 FZ -0.6
148. 7 TO 12 FZ 0.6
149. LOAD 7 LOADTYPE WIND  TITLE WL+Z
150. JOINT LOAD
151. 1 TO 6 FZ 1.6
152. 7 TO 12 FZ 9
153. 1 4 7 10 FX -10
154. 3 6 9 12 FX 10
155. LOAD 8 LOADTYPE WIND  TITLE WL-Z
156. JOINT LOAD
157. 1 TO 6 FZ 15
158. 7 TO 12 FZ -4
159. 1 4 7 10 FX -0.4
160. 3 6 9 12 FX 0.4
161. LOAD 9 LOADTYPE TEMPERATURE  TITLE TEMP
162. TEMPERATURE LOAD
163. 1 TO 11 13 25 TO 30 37 38 TEMP 25
164. ************************************************
165. LOAD COMB 100 SLS: (DL)
166. 3 1.0
167. LOAD COMB 101 SLS: (DL+T)
168. 3 1.0 9 1.0
169. LOAD COMB 102 SLS: (D+LR)
170. 3 1.0 4 1.0
171. LOAD COMB 103 SLS: (D+0.75LR)
172. 3 1.0 4 0.75
173. LOAD COMB 104 SLS: (D+0.75*0.7EZ)
174. 3 1.0 1 0.525
175. LOAD COMB 105 SLS: (D+0.75*0.7EX)
176. 3 1.0 2 0.525
177. LOAD COMB 106 SLS: (D+0.75WX1+0.75LR)
178. 3 1.0 5 0.75 4 0.75
179. LOAD COMB 107 SLS: (D+0.75WX2+0.75LR)
180. 3 1.0 6 0.75 4 0.75
181. LOAD COMB 108 SLS: (0.6D+WX1)
182. 3 0.6 5 1.0
183. LOAD COMB 109 SLS: (0.6D+WX2)
184. 3 0.6 6 1.0
185. ****************************************
186. LOAD COMB 110 SLS: (D+0.75WZ1+0.75LR)
187. 3 1.0 7 0.75 4 0.75
```

```
188. LOAD COMB 111 SLS: (D+0.75WZ2+0.75LR)
189. 3 1.0 8 0.75 4 0.75
190. LOAD COMB 112 SLS: (0.6D+WZ1)
191. 3 0.6 7 1.0
192. LOAD COMB 113 SLS: (0.6D+WZ2)
193. 3 0.6 8 1.0
194. ********************************************
195. LOAD COMB 114 SLS: (0.6D+0.7EX)
196. 3 0.6 2 0.7
197. LOAD COMB 115 SLS: (0.6D+0.7EZ)
198. 3 0.6 1 0.7
199. ************************************************
200. ******************************************************
201. LOAD COMB 200 ULS: (1.4 DL)
202. 3 1.4
203. LOAD COMB 201 ULS: (1.4D+1.4T(+VE)+0.5LR)
204. 3 1.4 4 0.5 9 1.4
205. LOAD COMB 202 ULS: (1.4D+1.4T(-VE)+0.5LR)
206. 3 1.4 4 0.5 9 -1.4
207. LOAD COMB 203 ULS: (1.2D+1.6WX1+0.5LR)
208. 3 1.2 5 1.6 4 0.5
209. LOAD COMB 204 ULS: (1.2D+1.6WX2+0.5LR)
210. 3 1.2 6 1.6 4 0.5
211. LOAD COMB 205 ULS: (1.2D+1.6WZ1+0.5LR)
212. 3 1.2 4 0.5 7 1.6
213. LOAD COMB 206 ULS: (1.2D+1.6WZ2+0.5LR)
214. 3 1.2 4 0.5 8 1.6
215. LOAD COMB 207 ULS: (1.2D+EZ)
216. 3 1.2 1 1.0
217. LOAD COMB 208 ULS: (1.2D+EX)
218. 3 1.2 2 1.0
219. LOAD COMB 209 ULS: (0.90D+1.6WX1)
220. 3 0.9 5 1.6
221. LOAD COMB 210 ULS: (0.90D+1.6WX2)
222. 3 0.9 6 1.6
223. LOAD COMB 211 ULS: (0.90D+1.6WZ1)
224. 3 0.9 7 1.6
225. LOAD COMB 212 ULS: (0.90D+1.6WZ2)
226. 3 0.9 8 1.6
227. LOAD COMB 213 ULS: (0.90D+EZ)
228. 3 0.9 1 1.0
229. LOAD COMB 214 ULS: (0.90D+EX)
230. 3 0.9 2 1.0
231. ************************************************************
232. PERFORM ANALYSIS PRINT STATICS CHECK
```

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE DEAD TITLE DL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.250000697E+01
Y = 0.159329920E+01
Z = 0.145000357E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -1820.53
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2639.77 MY= 0.00 MZ= -4551.34

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 1820.53
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2639.77 MY= 0.00 MZ= 4551.34

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	3.21188E-03	10
Y =	-2.73713E-02	11
Z =	4.29687E-03	33
RX=	2.37112E-04	5
RY=	4.41191E-07	6
RZ=	-1.90003E-04	7

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE LIVE TITLE LL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.250000000E+01
Y = 0.499983492E+01
Z = 0.145000009E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -50.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 73.89 MY= 0.00 MZ= -127.40

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 50.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -73.89 MY= 0.00 MZ= 127.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	5.30116E-04	31
Y =	-1.49011E-03	11
Z =	4.33051E-04	34
RX=	2.02440E-05	6
RY=	1.54726E-09	31
RZ=	-2.15573E-05	4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
LOADTYPE WIND TITLE WL+X

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.667469876E+01
Y = 0.249989076E+01
Z = 0.145000009E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 33.20
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.01 MY= 48.14 MZ= -83.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -33.20
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.01 MY= -48.14 MZ= 83.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	6.82500E-02	12
Y =	-3.37333E-04	12
Z =	-1.13050E-03	5
RX=	-1.11208E-05	5
RY=	9.29886E-08	4
RZ=	-9.57163E-05	31

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
LOADTYPE WIND TITLE WL-X

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 32.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 46.40 MZ= -79.99

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -32.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -46.40 MZ= 79.99

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	6.63337E-02	10
Y =	3.32090E-04	10
Z =	-4.86334E-05	5
RX=	-4.76752E-07	5
RY=	6.36495E-09	4
RZ=	-9.50676E-05	32

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
LOADTYPE WIND TITLE WL+Z

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.250000002E+01
Y = 0.249992522E+01
Z = 0.336981147E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 63.60

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 159.00 MY= -159.00 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -63.60

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -159.00 MY= 159.00 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	-1.39464E-03	10
Y =	-8.56527E-04	12
Z =	1.51671E-01	11
RX=	2.26553E-04	33
RY=	-2.80627E-06	10
RZ=	9.45101E-06	10

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
LOADTYPE WIND TITLE WL-Z

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 66.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 164.99 MY= -165.00 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -66.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -164.99 MY= 165.00 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X =	-6.24755E-05	10
Y =	-8.70736E-04	10
Z =	1.57851E-01	5
RX=	2.38404E-04	30
RY=	-2.94701E-06	4
RZ=	4.30461E-07	10

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
LOADTYPE TEMPERATURE TITLE TEMP

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 0.0000000E+00
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -1.2642606E-13

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3.2025601E-04 MY= 0.0000000E+00 MZ= -3.2025827E-04

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 1.7383584E-13
SUMMATION FORCE-Y = 7.7337194E-13
SUMMATION FORCE-Z = -5.5311403E-13

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.0911732E-10 MY= 1.7299495E-10 MZ= -2.8431172E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 9)

	MAXIMUMS	AT NODE
X =	-5.81629E-02	1
Y =	5.50295E-02	11
Z =	-3.12579E-02	3
RX=	-2.78654E-04	5
RY=	-2.38467E-06	6
RZ=	3.16605E-04	10

********** END OF DATA FROM INTERNAL STORAGE **********

233. LOAD LIST 100 TO 115
234. PRINT SUPPORT REACTION LIST 23 TO 28

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
23	100	24.49	254.98	27.67	16.14	-0.01	-13.74
	101	83.92	261.99	60.71	53.95	-0.02	-82.85
	102	24.31	261.97	27.47	16.04	-0.01	-13.67
	103	24.35	260.22	27.52	16.06	-0.01	-13.69
	104	24.48	250.03	24.03	8.29	0.11	-13.73
	105	21.18	250.62	27.67	16.14	-0.01	-6.14
	106	20.99	257.11	28.23	16.86	-0.01	-7.17
	107	20.22	257.09	27.55	16.10	-0.01	-6.38
	108	10.21	148.84	17.55	10.74	0.00	0.44
	109	9.18	148.81	16.64	9.73	0.00	1.50
	110	25.24	252.57	19.71	1.49	0.00	-14.71
	111	24.39	252.15	18.74	0.18	0.00	-13.73
	112	15.88	142.79	6.19	-9.76	0.00	-9.61
	113	14.74	142.22	4.89	-11.50	0.00	-8.30
	114	10.29	147.17	16.60	9.68	0.00	1.88
	115	14.68	146.38	11.75	-0.78	0.15	-8.23
24	100	0.00	400.30	26.81	15.70	0.00	0.00
	101	0.00	386.28	59.95	53.56	0.00	0.00
	102	0.00	411.80	26.62	15.60	0.00	0.00
	103	0.00	408.92	26.66	15.62	0.00	0.00
	104	0.00	394.74	22.76	6.70	0.00	0.00
	105	-5.53	400.30	26.81	15.70	0.00	8.92
	106	-5.17	408.84	27.38	16.42	0.00	7.67
	107	-4.98	409.11	26.69	15.66	0.00	7.39
	108	-6.89	240.07	17.04	10.48	0.00	10.22
	109	-6.64	240.43	16.13	9.46	0.00	9.85
	110	0.00	401.27	18.99	1.22	0.00	0.00
	111	0.00	401.25	18.02	-0.08	0.00	0.00
	112	0.00	229.98	5.86	-9.79	0.00	0.00
	113	0.00	229.95	4.56	-11.52	0.00	0.00
	114	-7.37	240.18	16.09	9.42	0.00	11.89
	115	0.00	232.76	10.69	-2.58	0.00	0.00
25	100	-24.49	254.98	27.67	16.14	0.01	13.74
	101	-83.92	261.99	60.71	53.95	0.02	82.85
	102	-24.31	261.97	27.47	16.04	0.01	13.67
	103	-24.35	260.23	27.52	16.06	0.01	13.69
	104	-24.48	250.03	24.03	8.29	-0.11	13.73
	105	-27.79	259.35	27.67	16.14	0.01	21.34
	106	-28.27	263.42	28.23	16.86	0.01	20.85
	107	-27.24	263.17	27.55	16.10	0.01	19.56
	108	-19.92	157.25	17.55	10.74	0.00	17.79
	109	-18.54	156.92	16.64	9.73	0.00	16.08
	110	-25.24	252.57	19.71	1.49	0.00	14.71
	111	-24.39	252.15	18.74	0.18	0.00	13.73
	112	-15.88	142.79	6.19	-9.75	0.00	9.61
	113	-14.74	142.23	4.89	-11.50	0.00	8.30

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	114	-19.10	158.81	16.60	9.68	0.00	18.37
	115	-14.68	146.38	11.75	-0.78	-0.15	8.23
26	100	24.49	254.98	-27.67	-16.14	0.01	-13.74
	101	83.92	261.99	-60.71	-53.95	0.02	-82.85
	102	24.31	261.97	-27.47	-16.04	0.01	-13.67
	103	24.35	260.22	-27.52	-16.06	0.01	-13.69
	104	24.49	259.94	-31.30	-23.99	0.12	-13.75
	105	21.18	250.62	-27.67	-16.14	0.01	-6.14
	106	20.99	257.11	-28.23	-16.86	0.01	-7.17
	107	20.22	257.09	-27.55	-16.10	0.01	-6.38
	108	10.21	148.84	-17.55	-10.74	0.00	0.44
	109	9.18	148.81	-16.64	-9.73	0.00	1.50
	110	25.24	268.14	-35.71	-31.06	0.01	-14.71
	111	24.39	268.31	-35.34	-30.87	0.01	-13.73
	112	15.88	163.55	-27.51	-29.68	0.01	-9.61
	113	14.74	163.77	-27.02	-29.43	0.01	-8.30
	114	10.29	147.17	-16.60	-9.68	0.00	1.88
	115	14.70	159.60	-21.45	-20.15	0.16	-8.25
27	100	0.00	400.30	-26.81	-15.70	0.00	0.00
	101	0.00	386.28	-59.95	-53.56	0.00	0.00
	102	0.00	411.80	-26.62	-15.60	0.00	0.00
	103	0.00	408.93	-26.66	-15.62	0.00	0.00
	104	0.00	405.86	-30.85	-24.70	0.00	0.00
	105	-5.53	400.30	-26.81	-15.70	0.00	8.92
	106	-5.17	408.84	-27.38	-16.42	0.00	7.67
	107	-4.98	409.11	-26.69	-15.66	0.00	7.39
	108	-6.89	240.07	-17.04	-10.48	0.00	10.22
	109	-6.64	240.43	-16.13	-9.46	0.00	9.85
	110	0.00	416.04	-34.71	-30.45	0.00	0.00
	111	0.00	416.58	-34.33	-30.25	0.00	0.00
	112	0.00	249.67	-26.82	-29.19	0.00	0.00
	113	0.00	250.38	-26.31	-28.92	0.00	0.00
	114	-7.37	240.18	-16.09	-9.42	0.00	11.89
	115	0.00	247.59	-21.48	-21.42	0.00	0.00
28	100	-24.49	254.98	-27.67	-16.14	-0.01	13.74
	101	-83.92	262.00	-60.71	-53.95	-0.02	82.85
	102	-24.31	261.97	-27.47	-16.04	-0.01	13.67
	103	-24.35	260.23	-27.52	-16.06	-0.01	13.69
	104	-24.49	259.94	-31.30	-23.99	-0.12	13.75
	105	-27.79	259.35	-27.67	-16.14	-0.01	21.34
	106	-28.27	263.42	-28.23	-16.86	-0.01	20.85
	107	-27.24	263.17	-27.55	-16.10	-0.01	19.56
	108	-19.92	157.25	-17.55	-10.74	0.00	17.79
	109	-18.54	156.92	-16.64	-9.73	0.00	16.08
	110	-25.24	268.15	-35.71	-31.06	-0.01	14.71
	111	-24.39	268.31	-35.34	-30.87	-0.01	13.73

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	112	-15.88	163.55	-27.51	-29.68	-0.01	9.61
	113	-14.74	163.77	-27.02	-29.43	-0.01	8.30
	114	-19.10	158.81	-16.60	-9.68	0.00	18.37
	115	-14.70	159.60	-21.45	-20.15	-0.16	8.25

********** END OF LATEST ANALYSIS RESULT **********

235. LOAD LIST 200 TO 214
236. PRINT SUPPORT REACTION LIST 23 TO 28

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
23	200	34.28	356.98	38.73	22.59	-0.01	-19.23
	201	117.40	370.29	84.89	75.48	-0.03	-115.96
	202	-49.03	350.66	-7.62	-30.39	0.01	77.56
	203	22.12	302.84	34.62	21.01	-0.01	-2.55
	204	20.47	302.79	33.17	19.39	-0.01	-0.86
	205	31.20	293.15	16.45	-11.78	0.00	-18.64
	206	29.37	292.25	14.37	-14.58	0.01	-16.54
	207	29.37	296.54	26.28	4.41	0.21	-16.47
	208	23.09	297.66	33.20	19.37	-0.01	-2.02
	209	14.87	222.85	26.42	16.22	-0.01	1.53
	210	13.22	222.80	24.96	14.60	-0.01	3.22
	211	23.94	213.16	8.24	-16.58	0.01	-14.55
	212	22.11	212.26	6.17	-19.37	0.01	-12.45
	213	22.03	220.04	17.98	-0.43	0.22	-12.35
	214	15.74	221.17	24.90	14.52	-0.01	2.10
24	200	0.00	560.42	37.53	21.98	0.00	0.00
	201	0.00	546.54	83.83	74.93	0.00	0.00
	202	0.00	585.79	-8.96	-31.07	0.00	0.00
	203	-11.02	485.93	33.60	20.49	0.00	16.36
	204	-10.62	486.50	32.14	18.86	0.00	15.76
	205	0.00	469.79	15.71	-11.94	0.00	0.00
	206	0.00	469.74	13.64	-14.72	0.00	0.00
	207	0.00	469.76	24.47	1.69	0.00	0.00
	208	-10.53	480.36	32.17	18.84	0.00	16.98
	209	-11.02	360.09	25.65	15.83	0.00	16.36
	210	-10.62	360.66	24.19	14.20	0.00	15.76
	211	0.00	343.95	7.77	-16.60	0.00	0.00
	212	0.00	343.90	5.70	-19.38	0.00	0.00
	213	0.00	349.68	16.42	-3.02	0.00	0.00
	214	-10.53	360.27	24.13	14.13	0.00	16.98
25	200	-34.28	356.98	38.73	22.59	0.01	19.24
	201	-117.40	370.28	84.89	75.48	0.03	115.96
	202	49.02	350.66	-7.62	-30.39	-0.01	-77.56
	203	-37.66	316.29	34.62	21.01	0.01	31.73
	204	-35.45	315.76	33.17	19.39	0.01	28.98
	205	-31.20	293.15	16.45	-11.78	0.00	18.64
	206	-29.37	292.25	14.37	-14.57	-0.01	16.54
	207	-29.37	296.54	26.28	4.41	-0.21	16.47
	208	-35.68	314.30	33.20	19.37	0.01	30.96
	209	-30.41	236.30	26.42	16.22	0.01	27.64
	210	-28.19	235.77	24.96	14.60	0.01	24.90
	211	-23.94	213.16	8.24	-16.58	-0.01	14.55
	212	-22.11	212.26	6.17	-19.37	-0.01	12.45
	213	-22.03	220.04	17.98	-0.43	-0.22	12.35
	214	-28.33	237.80	24.90	14.53	0.01	26.83
26	200	34.28	356.98	-38.73	-22.59	0.01	-19.24

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	201	117.40	370.29	-84.89	-75.48	0.03	-115.96
	202	-49.02	350.66	7.62	30.39	-0.01	77.56
	203	22.12	302.84	-34.62	-21.01	0.01	-2.55
	204	20.47	302.79	-33.17	-19.39	0.01	-0.87
	205	31.19	326.37	-50.56	-51.32	0.02	-18.64
	206	29.37	326.72	-49.77	-50.91	0.02	-16.54
	207	29.39	315.42	-40.12	-34.32	0.23	-16.50
	208	23.09	297.66	-33.20	-19.37	0.01	-2.02
	209	14.87	222.85	-26.42	-16.22	0.01	1.53
	210	13.22	222.80	-24.96	-14.60	0.01	3.22
	211	23.94	246.38	-42.36	-46.52	0.02	-14.55
	212	22.11	246.73	-41.57	-46.11	0.02	-12.45
	213	22.05	238.93	-31.82	-29.48	0.23	-12.38
	214	15.74	221.17	-24.90	-14.53	0.01	2.10
27	200	0.00	560.42	-37.53	-21.98	0.00	0.00
	201	0.00	546.54	-83.83	-74.93	0.00	0.00
	202	0.00	585.80	8.96	31.07	0.00	0.00
	203	-11.02	485.93	-33.60	-20.49	0.00	16.36
	204	-10.62	486.51	-32.14	-18.86	0.00	15.76
	205	0.00	501.29	-49.24	-50.41	0.00	0.00
	206	0.00	502.44	-48.44	-49.99	0.00	0.00
	207	0.00	490.95	-39.87	-35.98	0.00	0.00
	208	-10.53	480.36	-32.17	-18.84	0.00	16.98
	209	-11.02	360.09	-25.65	-15.83	0.00	16.36
	210	-10.62	360.67	-24.19	-14.20	0.00	15.76
	211	0.00	375.45	-41.30	-45.75	0.00	0.00
	212	0.00	376.59	-40.49	-45.33	0.00	0.00
	213	0.00	370.86	-31.83	-31.27	0.00	0.00
	214	-10.53	360.27	-24.13	-14.13	0.00	16.98
28	200	-34.28	356.98	-38.73	-22.59	-0.01	19.24
	201	-117.40	370.29	-84.90	-75.48	-0.03	115.96
	202	49.02	350.66	7.62	30.39	0.01	-77.56
	203	-37.66	316.29	-34.62	-21.01	-0.01	31.73
	204	-35.45	315.76	-33.17	-19.39	-0.01	28.98
	205	-31.19	326.37	-50.56	-51.32	-0.02	18.64
	206	-29.37	326.72	-49.77	-50.91	-0.02	16.54
	207	-29.39	315.42	-40.12	-34.32	-0.23	16.50
	208	-35.68	314.30	-33.20	-19.37	-0.01	30.96
	209	-30.41	236.30	-26.42	-16.22	-0.01	27.64
	210	-28.19	235.77	-24.96	-14.60	-0.01	24.90
	211	-23.94	246.38	-42.36	-46.52	-0.02	14.55
	212	-22.11	246.73	-41.57	-46.11	-0.02	12.45
	213	-22.05	238.93	-31.82	-29.48	-0.23	12.38
	214	-28.33	237.80	-24.90	-14.53	-0.01	26.84

********** END OF LATEST ANALYSIS RESULT **********

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237. LOAD LIST 200 TO 214
238. START CONCRETE DESIGN
239. CODE ACI
240. FC 35000 ALL
241. EFACE 0.44 MEMB 1 2 6 7 41 43 46 48
242. SFACE 0.44 MEMB 1 2 6 7 41 43 46 48
243. EFACE 0.29 MEMB 11 13 49 50
244. SFACE 0.29 MEMB 11 13 49 50
245. CLB 0.04 MEMB 1 TO 11 13 27 30 TO 36 39 TO 50
246. CLS 0.04 MEMB 1 TO 11 13 27 30 TO 36 39 TO 50
247. CLT 0.04 MEMB 1 TO 11 13 27 30 TO 36 39 TO 50
248. MAXMAIN 16 MEMB 1 TO 11 13 27 30 TO 36 39 TO 50
249. MINMAIN 16 MEMB 1 TO 11 13 27 30 TO 36 39 TO 50
250. MINMAIN 20 MEMB 4 9 32 35 40 45
251. MAXMAIN 20 MEMB 4 9 32 35 40 45
252. MINSEC 10 MEMB 1 TO 11 13 27 30 TO 36 39 TO 50
253. DESIGN BEAM 1 2 6 7 11 13 25 TO 30 37 38 41 43 46 48 TO 51
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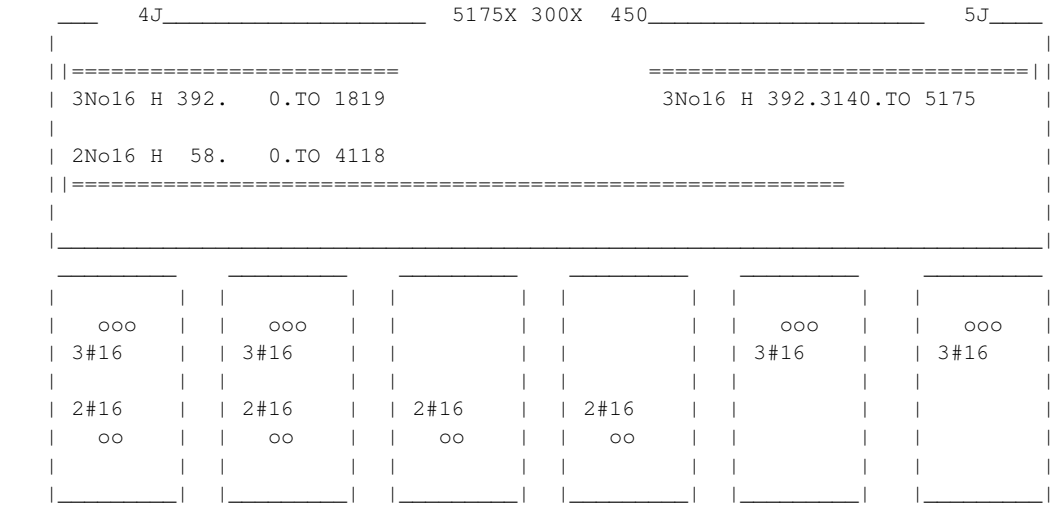
BEAM NO. 1 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4118.	YES	NO
2	392.	3 - 16MM	0.	1819.	YES	NO
3	392.	3 - 16MM	3140.	5175.	NO	YES

B E A M N O. 1 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 25.08 KNS Vc= 207.65 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.51 KNS Vc= 123.98 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



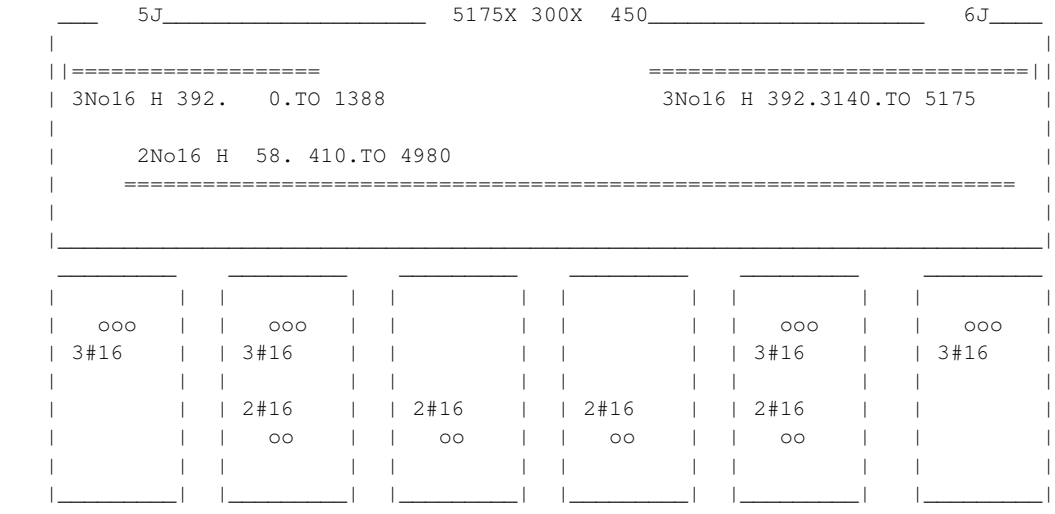
BEAM NO. 2 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	410.	4980.	NO	NO
2	392.	3 - 16MM	0.	1388.	YES	NO
3	392.	3 - 16MM	3140.	5175.	NO	YES

B E A M N O. 2 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 28.50 KNS Vc= 123.98 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 25.08 KNS Vc= 207.65 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



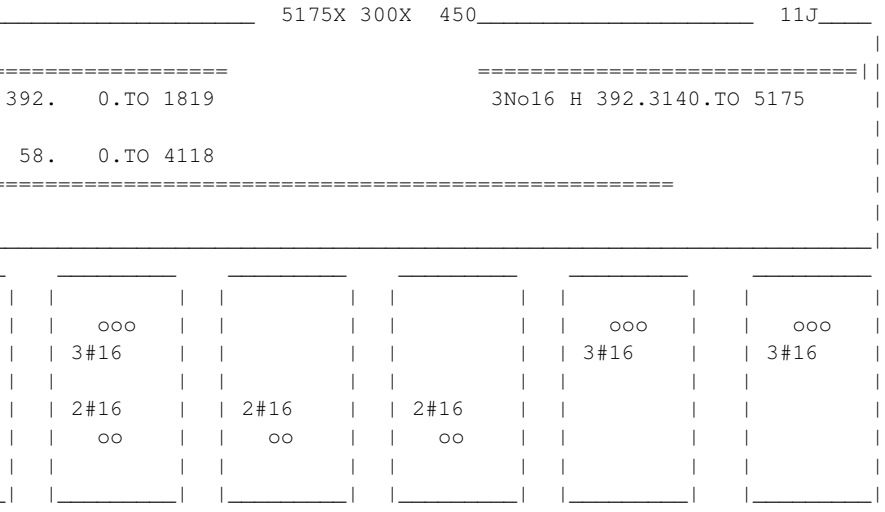
BEAM NO. 6 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4118.	YES	NO
2	392.	3 - 16MM	0.	1819.	YES	NO
3	392.	3 - 16MM	3140.	5175.	NO	YES

B E A M N O. 6 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 25.08 KNS Vc= 207.65 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.51 KNS Vc= 123.98 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



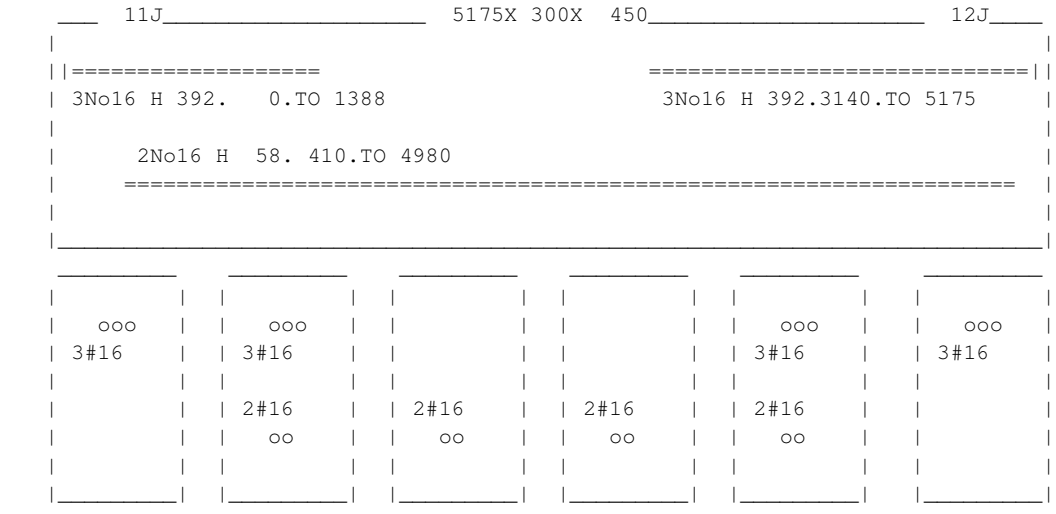
BEAM NO. 7 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	410.	4980.	NO	NO
2	392.	3 - 16MM	0.	1388.	YES	NO
3	392.	3 - 16MM	3140.	5175.	NO	YES

B E A M N O. 7 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 28.50 KNS Vc= 123.98 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 25.08 KNS Vc= 207.65 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



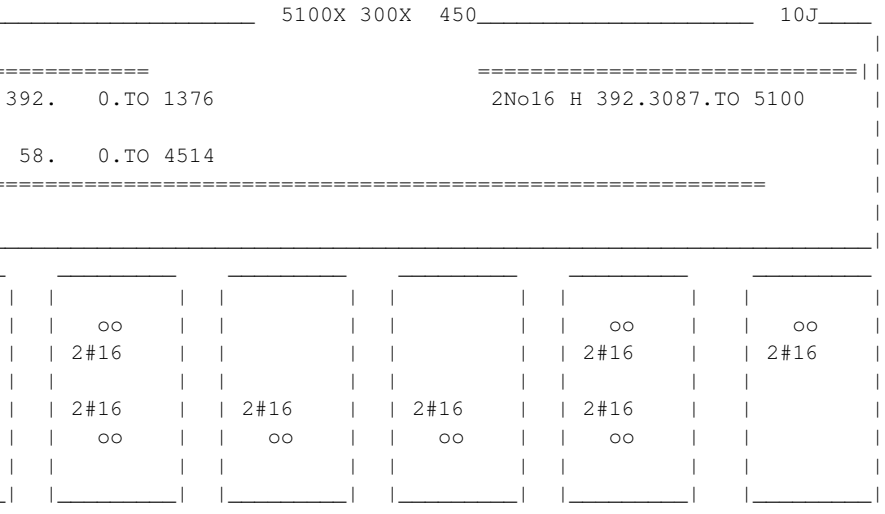
BEAM NO. 11 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5100. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4514.	YES	NO
2	392.	2 - 16MM	0.	1376.	YES	NO
3	392.	2 - 16MM	3087.	5100.	NO	YES

B E A M N O. 11 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 28.51 KNS Vc= 209.86 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.51 KNS Vc= 209.86 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



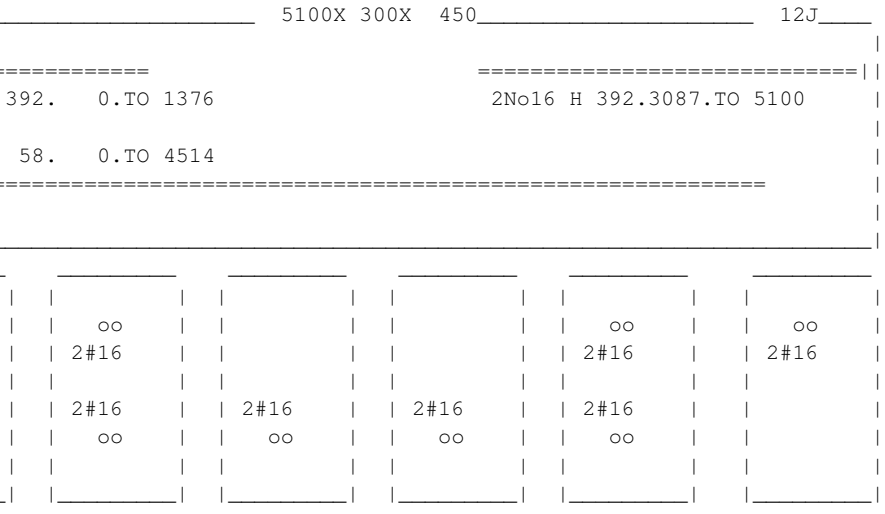
BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5100. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4514.	YES	NO
2	392.	2 - 16MM	0.	1376.	YES	NO
3	392.	2 - 16MM	3087.	5100.	NO	YES

B E A M N O. 13 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 28.50 KNS Vc= 209.86 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.51 KNS Vc= 209.86 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 25 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

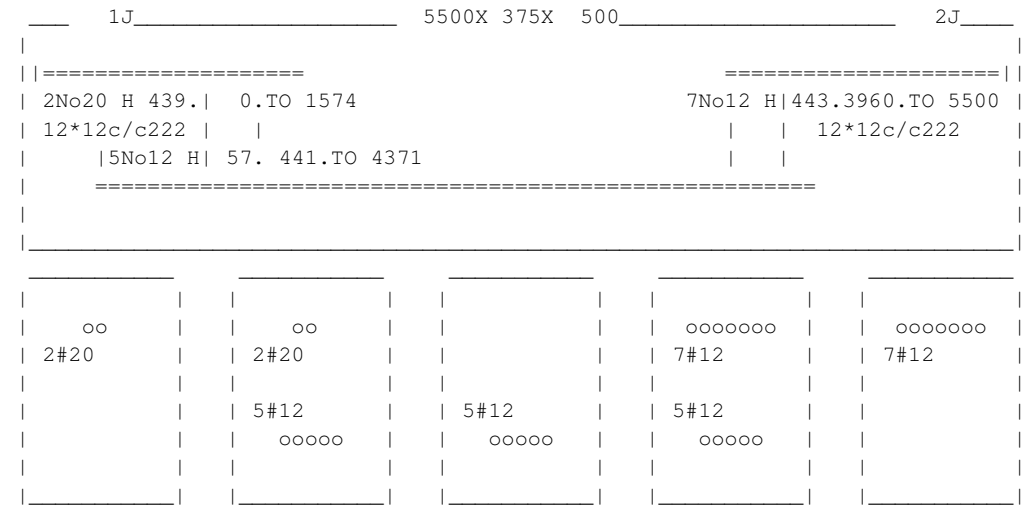
LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 375. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	441.	4371.	NO	NO
2	439.	2 - 20MM	0.	1574.	YES	NO
3	443.	7 - 12MM	3960.	5500.	NO	YES

B E A M N O. 25 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 84.91 KNS Vc= 163.40 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2313. MM

AT END SUPPORT - Vu= 92.76 KNS Vc= 161.26 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2313. MM



BEAM NO. 26 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

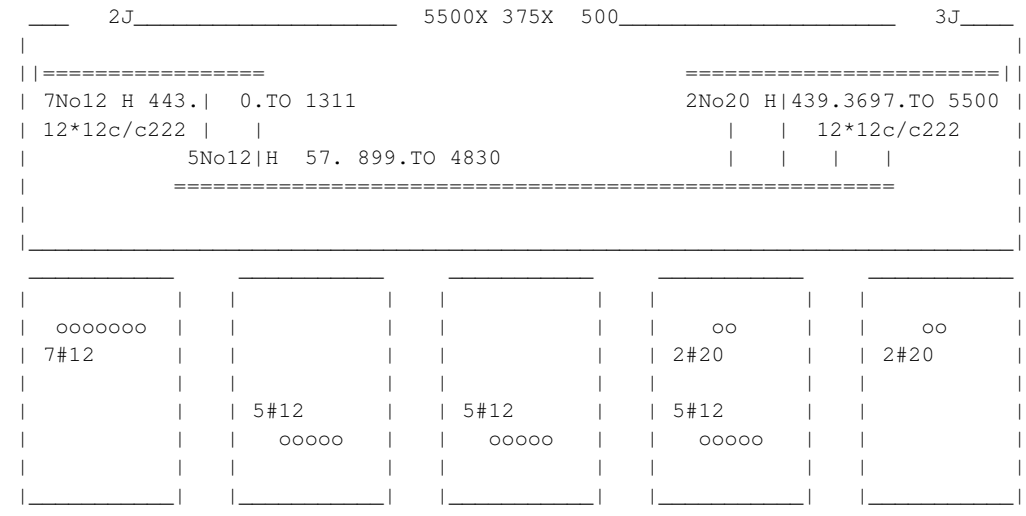
LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 375. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	899.	4830.	NO	NO
2	443.	7 - 12MM	0.	1311.	YES	NO
3	439.	2 - 20MM	3697.	5500.	NO	YES

B E A M N O. 26 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 92.76 KNS Vc= 161.26 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2313. MM

AT END SUPPORT - Vu= 84.91 KNS Vc= 163.40 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2313. MM



BEAM NO. 27 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

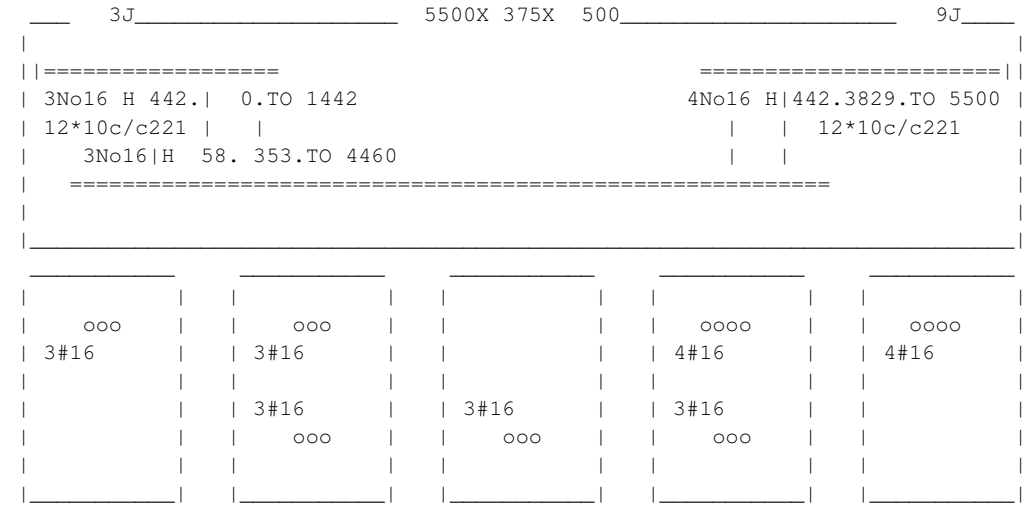
LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 375. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	3 - 16MM	353.	4460.	NO	NO
2	442.	3 - 16MM	0.	1442.	YES	NO
3	442.	4 - 16MM	3829.	5500.	NO	YES

B E A M N O. 27 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 88.70 KNS Vc= 163.08 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.7 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2310. MM

AT END SUPPORT - Vu= 88.70 KNS Vc= 163.08 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.7 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2310. MM



BEAM NO. 28 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

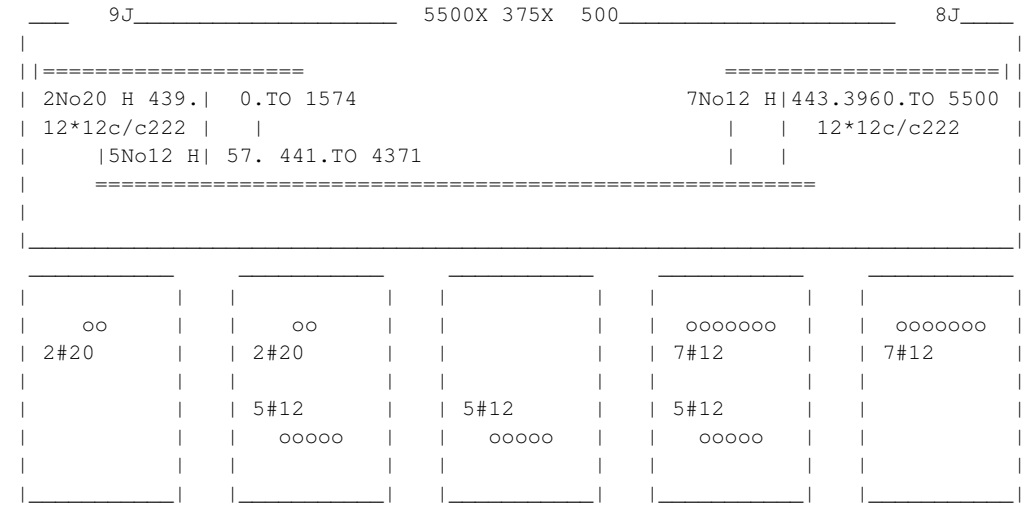
LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 375. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	441.	4371.	NO	NO
2	439.	2 - 20MM	0.	1574.	YES	NO
3	443.	7 - 12MM	3960.	5500.	NO	YES

B E A M N O. 28 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 84.91 KNS Vc= 163.40 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2313. MM

AT END SUPPORT - Vu= 92.76 KNS Vc= 161.26 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2313. MM



BEAM NO. 29 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

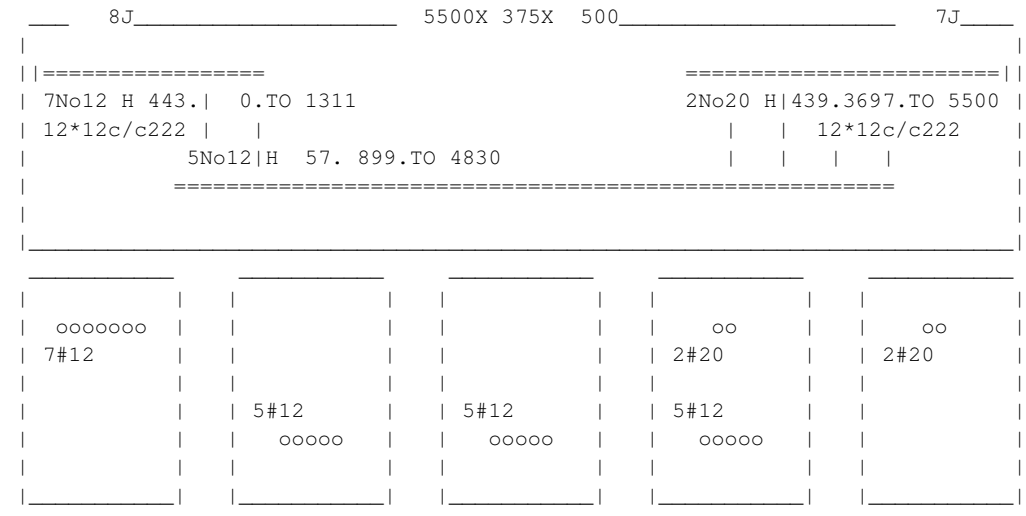
LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 375. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	899.	4830.	NO	NO
2	443.	7 - 12MM	0.	1311.	YES	NO
3	439.	2 - 20MM	3697.	5500.	NO	YES

B E A M N O. 29 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 92.76 KNS Vc= 161.26 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2313. MM

AT END SUPPORT - Vu= 84.91 KNS Vc= 163.40 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2313. MM



BEAM NO. 30 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

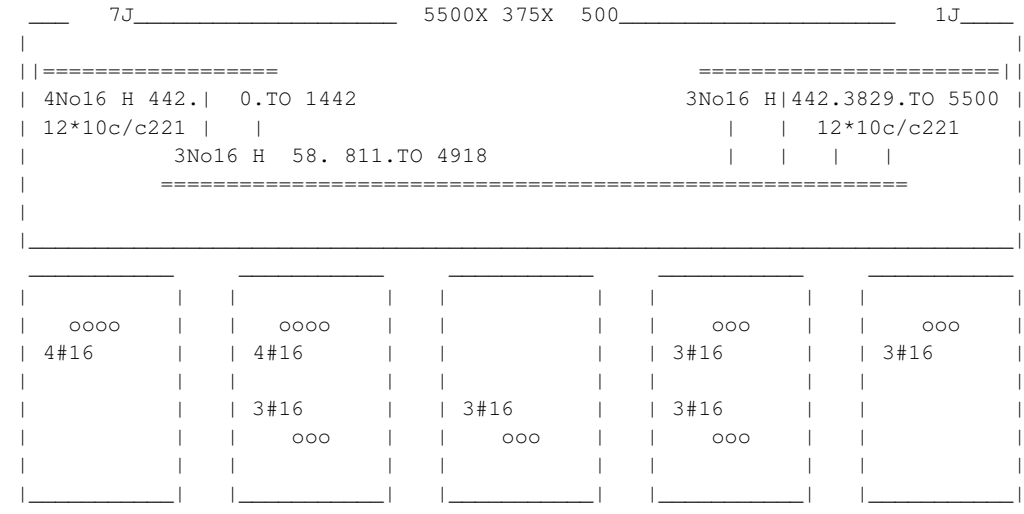
LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 375. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	3 - 16MM	811.	4918.	NO	NO
2	442.	4 - 16MM	0.	1442.	YES	NO
3	442.	3 - 16MM	3829.	5500.	NO	YES

B E A M N O. 30 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 88.70 KNS Vc= 163.08 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.7 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2310. MM

AT END SUPPORT - Vu= 88.70 KNS Vc= 163.08 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.7 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2310. MM



BEAM NO. 37 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

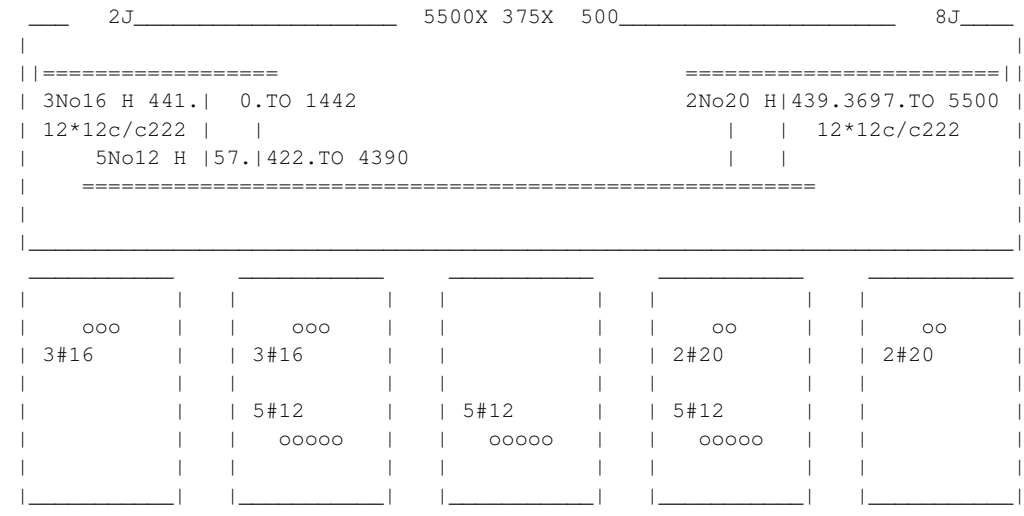
LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 375. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	422.	4390.	NO	NO
2	441.	3 - 16MM	0.	1442.	YES	NO
3	439.	2 - 20MM	3697.	5500.	NO	YES

B E A M N O. 37 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 88.84 KNS Vc= 163.47 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2314. MM

AT END SUPPORT - Vu= 88.84 KNS Vc= 163.47 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 7.6 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2314. MM



BEAM NO. 38 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

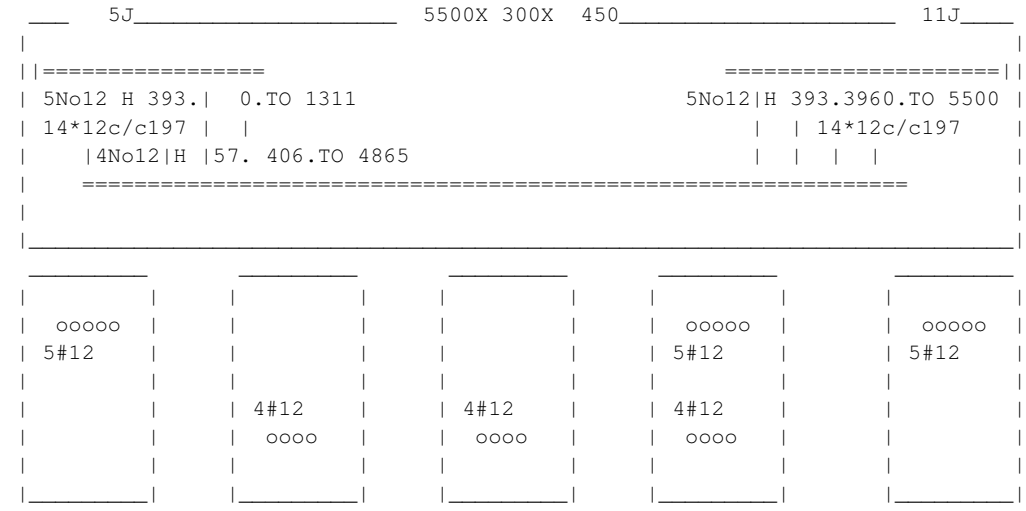
LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	406.	4865.	NO	NO
2	393.	5 - 12MM	0.	1311.	YES	NO
3	393.	5 - 12MM	3960.	5500.	NO	YES

B E A M N O. 38 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 61.84 KNS Vc= 117.50 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 197. MM C/C FOR 2364. MM

AT END SUPPORT - Vu= 61.84 KNS Vc= 117.50 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 200
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 197. MM C/C FOR 2364. MM



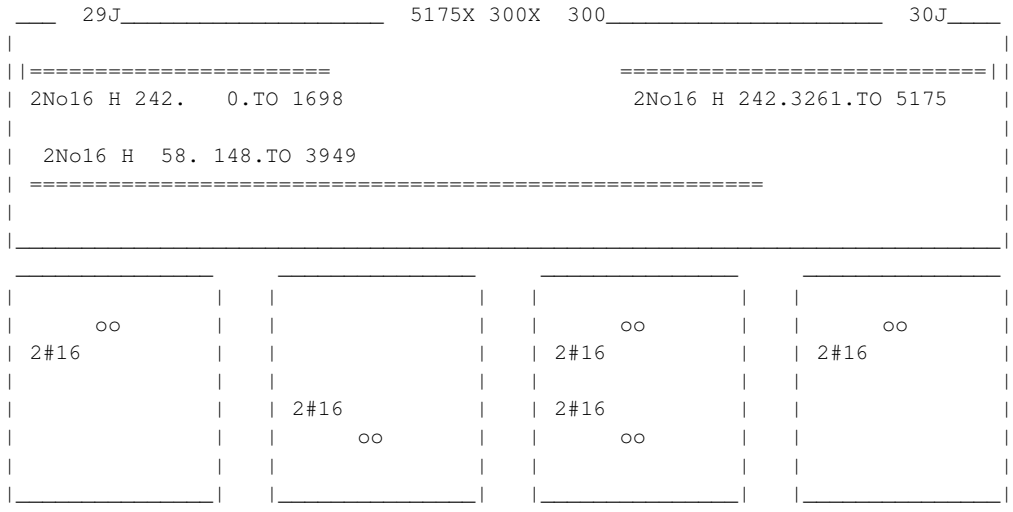
BEAM NO. 41 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	148.	3949.	NO	NO
2	242.	2 - 16MM	0.	1698.	YES	NO
3	242.	2 - 16MM	3261.	5175.	NO	YES

B E A M N O. 41 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.64 KNS Vc= 68.39 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.71 KNS Vc= 68.39 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



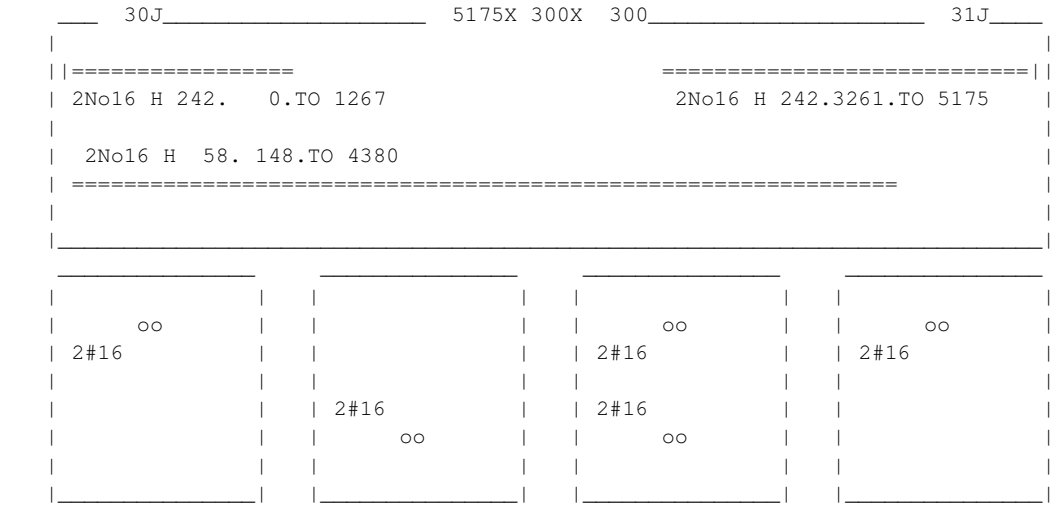
BEAM NO. 43 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	148.	4380.	NO	NO
2	242.	2 - 16MM	0.	1267.	YES	NO
3	242.	2 - 16MM	3261.	5175.	NO	YES

B E A M N O. 43 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.71 KNS Vc= 68.39 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.64 KNS Vc= 68.39 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



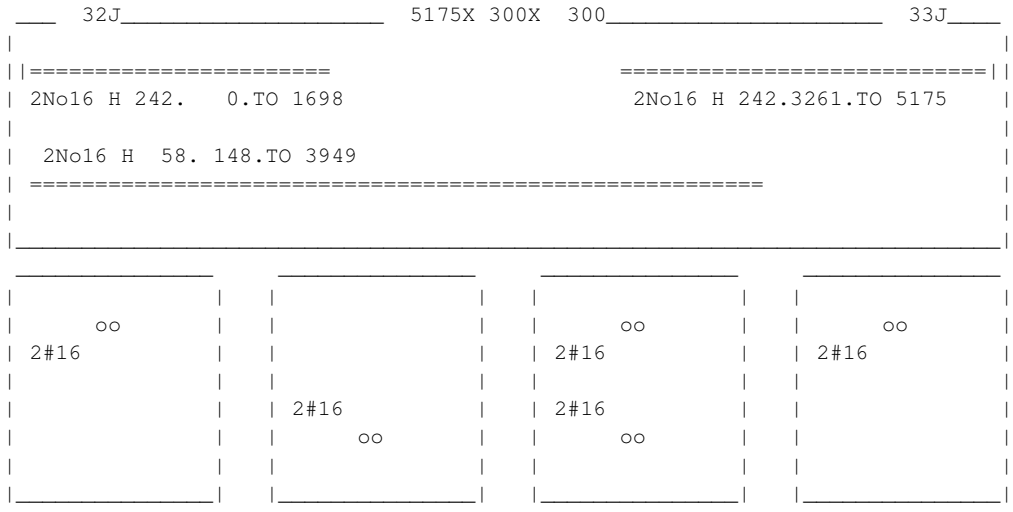
BEAM NO. 46 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	148.	3949.	NO	NO
2	242.	2 - 16MM	0.	1698.	YES	NO
3	242.	2 - 16MM	3261.	5175.	NO	YES

B E A M N O. 46 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.64 KNS Vc= 68.39 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.71 KNS Vc= 68.39 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



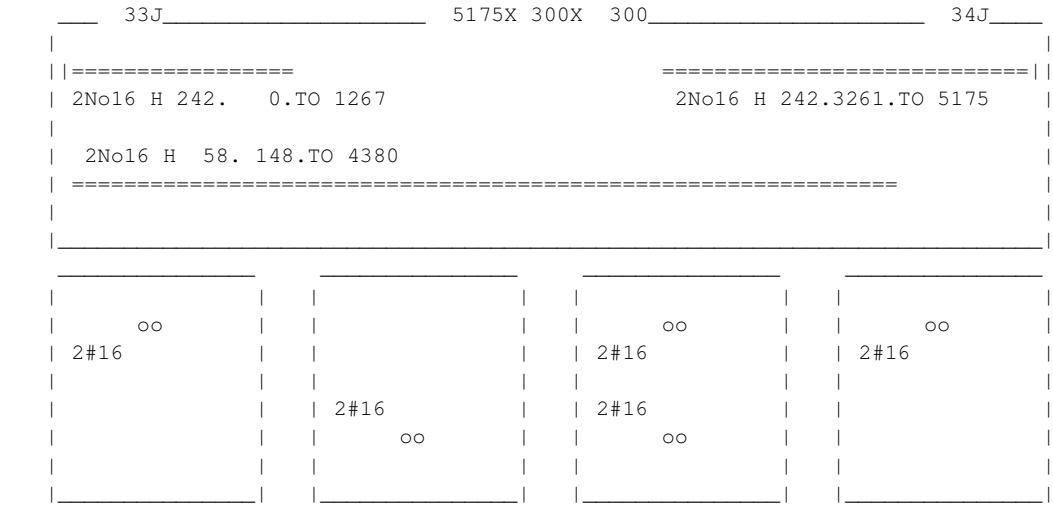
BEAM NO. 48 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	148.	4380.	NO	NO
2	242.	2 - 16MM	0.	1267.	YES	NO
3	242.	2 - 16MM	3261.	5175.	NO	YES

B E A M N O. 48 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.71 KNS Vc= 68.39 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.64 KNS Vc= 68.39 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 2.4 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



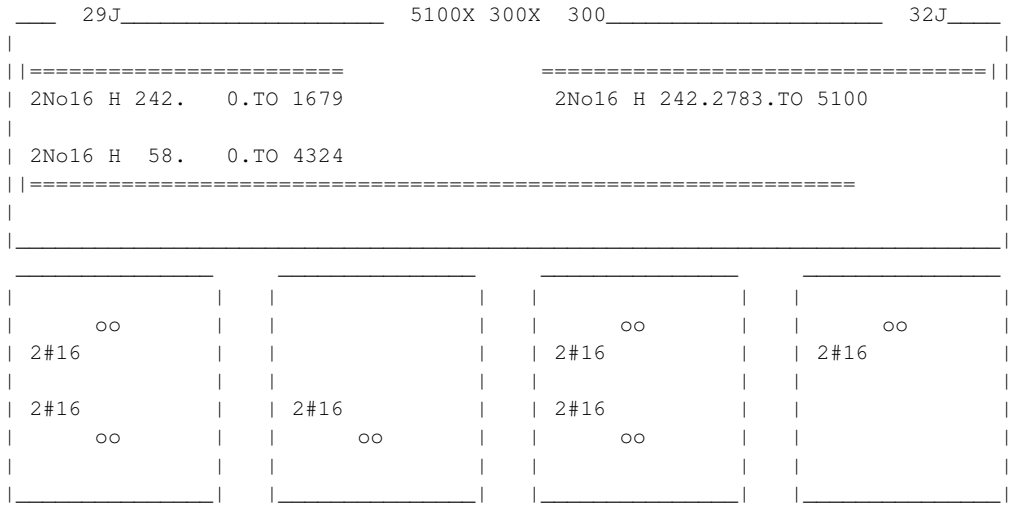
BEAM NO. 49 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5100. MM FY - 414. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4324.	YES	NO
2	242.	2 - 16MM	0.	1679.	YES	NO
3	242.	2 - 16MM	2783.	5100.	NO	YES

B E A M N O. 49 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.01 KNS Vc= 65.92 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.3 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.01 KNS Vc= 65.92 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.3 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



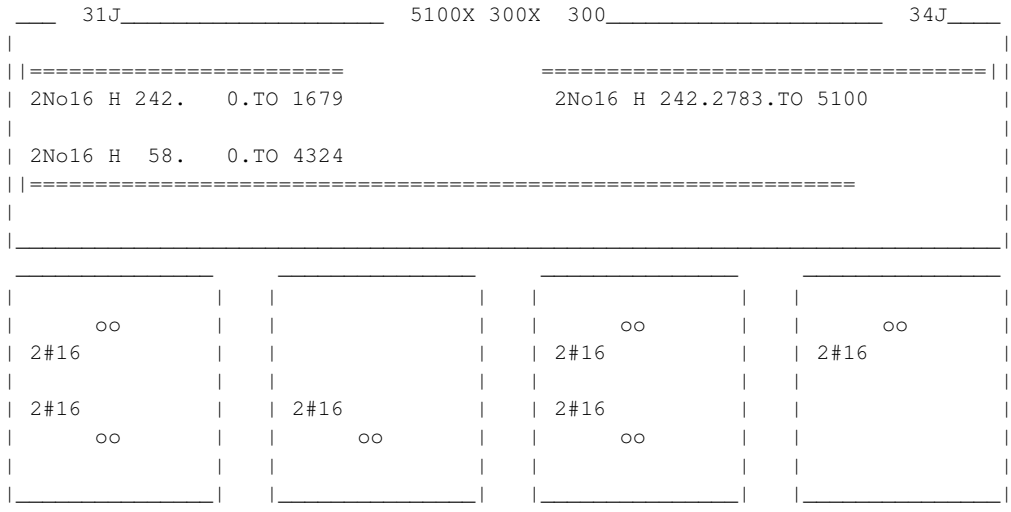
BEAM NO. 50 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5100. MM FY - 414. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4324.	YES	NO
2	242.	2 - 16MM	0.	1679.	YES	NO
3	242.	2 - 16MM	2783.	5100.	NO	YES

B E A M N O. 50 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.01 KNS Vc= 65.92 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.3 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.01 KNS Vc= 65.92 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.3 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



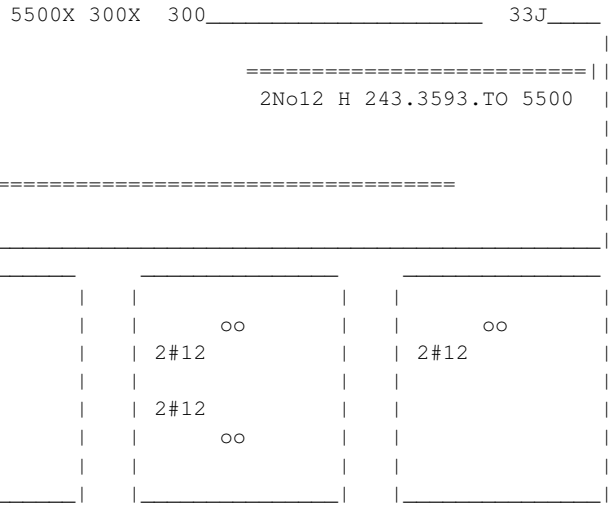
BEAM NO. 51 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 414. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	4696.	YES	NO
2	243.	2 - 12MM	0.	1220.	YES	NO
3	243.	2 - 12MM	3593.	5500.	NO	YES

B E A M N O. 51 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.48 KNS Vc= 61.99 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.2 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 7.48 KNS Vc= 61.99 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 2.2 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

254. DESIGN COLUMN 3 TO 5 8 TO 10 31 TO 36 39 40 42 44 45 47

==

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==

COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==========

COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

COLUMN NO. 31 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

COLUMN NO. 32 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==

COLUMN NO. 33 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==

COLUMN NO. 34 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==

COLUMN NO. 35 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==========

COLUMN NO. 36 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

COLUMN NO. 39 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

COLUMN NO. 40 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==========

COLUMN NO. 42 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

COLUMN NO. 44 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==

COLUMN NO. 45 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==

COLUMN NO. 47 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.287	200	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

255. END CONCRETE DESIGN
256. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 15,2020 TIME= 12:37:29 ****

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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	15-10-2020	31-10-2020	26-10-2023				Date	15-10-2020	31-10-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 2 Civil Foundation Input for Power Plant Station Transformer

CIVIL FOUNDATION INPUT FOR POWER PLANT STATION TRANSFORMER

